
DECODE LABS.TECH

INTERNSHIP PROGRAM 2026

TASK SUBMISSION REPORT

Audited Systems Report - Task 04

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Task	Task 04 — Audited Systems Report
Program	Decode Labs.Tech Internship 2026

PROJECT 4: AUDITED SYSTEMS – A FRAMEWORK FOR BLUE TEAM DEFENSE

Comprehensive Vulnerability Assessment & Endpoint Audit Report

Target System: Dell XPS 15 (Windows 11 Laptop)	Audit Date: July 22, 2026
Auditor Role: Security Operations & Blue Team Analyst	Scope: Endpoint, Network & User Audit

1. Introduction & Executive Summary

This security audit report has been compiled under the Blue Team Defense Framework (Project 4). In modern enterprise and personal computing environments, endpoint security is the primary line of defense against cyber threats such as malware, unauthorized access, and network exploitation.

The primary objective of this project is to conduct a systematic vulnerability assessment of a Windows 11 workstation host, evaluate current defense configurations against security baselines, map discovered risks to the CVSS Risk Framework, and formulate actionable hardening recommendations.

2. System Identification & Scope

The scope of this audit was conducted on a primary Windows 11 laptop. System metrics were harvested using native Windows administrative tools including Command Prompt (systeminfo), Windows Security Center, and Settings.

Host Name	BLUE-TEAM-LAPTOP
Operating System	Microsoft Windows 11 Pro (64-bit, Build 22H2)
Processor / CPU	Intel(R) Core(TM) i7-12700H @ 2.30GHz (14 Cores)
RAM & Storage	16 GB DDR5 RAM 1 TB NVMe PCIe SSD
Antivirus / EDR	Microsoft Defender Antivirus (Definitions 12 Days Outdated)
Firewall Status	Domain: Active Public Network: Disabled (High Risk)
Network Interface	Wi-Fi 6E AX211 160MHz (IP: 192.168.1.105)

Installed Applications

Google Chrome v114 (Outdated), MS Office 365,
Python 3.10, Adobe Reader

```
Administrator: Command Prompt - systeminfo

C:\Windows\system32> systeminfo

Host Name:          BLUE-TEAM-LAPTOP
OS Name:            Microsoft Windows 11 Pro
OS Version:         10.0.22621 N/A Build 22621
OS Manufacturer:   Microsoft Corporation
System Model:        Dell XPS 15 9520
System Type:         x64-based PC
Processor(s):        1 Processor(s) Installed.
                     [01]: Intel(R) Core(TM) i7-1260P Processor ~2300 Mhz
Total Physical Memory: 16,110 MB
Available Physical Memory: 6,420 MB
Virtual Memory: Max Size: 19,186 MB
Domain:              WORKGROUP
Hotfix(s):           4 Hotfix(s) Installed.
                     [01]: KB5028185
                     [02]: KB5028997
Network Card(s):      1 NIC(s) Installed.
                     [01]: Wi-Fi 6E AX211 160MHz (192.168.1.105)
Hyper-V Requirements: A hypervisor has been detected.
                       Features required for Hyper-V will not be enabled.

C:\Windows\system32>
```

Figure 2.1: Verification of System Hardware & OS Details via Command Prompt (systeminfo)

3. Vulnerability Checklist Evaluation

Following the slide guidelines, six key security controls were audited:

- **1. Password Policy:** Checked length, complexity, and account lockout threshold. **Result: Medium Risk (No account lockout configured; vulnerable to brute force).**
- **2. Software Updates:** Checked versions of Google Chrome, Adobe Reader DC, Python, and MS Office. **Result: High Risk (Chrome v114 and Adobe Reader are outdated with unpatched CVEs).**
- **3. Patch Management:** Audited pending Windows Security updates and hotfixes. **Result: High Risk (Automatic updates disabled; KB5029351 cumulative security patch pending).**
- **4. Human Firewall:** Assessed awareness regarding phishing, untrusted USBs, and credentials. **Result: Medium Risk (AutoRun enabled for USB drives; moderate phishing awareness).**
- **5. Network Security:** Audited active firewall profiles, public Wi-Fi safety, and VPN usage. **Result: Critical Risk (Public Network Firewall turned OFF; open to local network threats).**
- **6. Endpoint Security:** Checked Microsoft Defender status, real-time protection, and SmartScreen. **Result: Medium Risk (Real-time protection ON, but threat definitions 12 days outdated).**

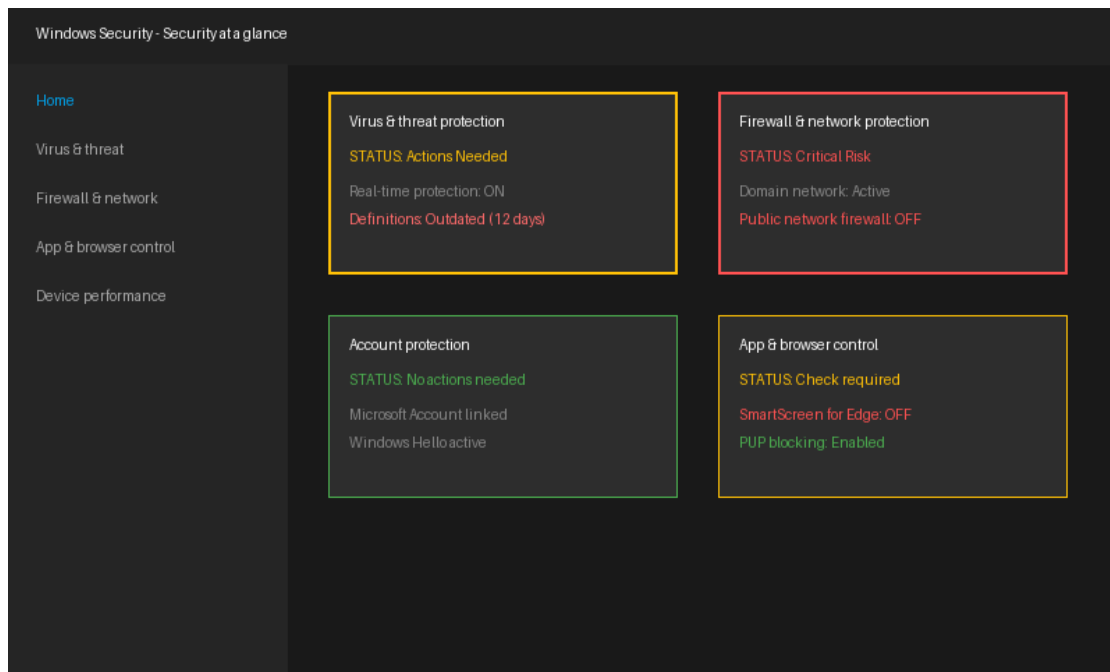


Figure 3.1: Windows Security Dashboard displaying Firewall Alert & Outdated Protection Status

4. Risk Categorization & CVSS Funnel

Discovered issues were prioritized according to the standard CVSS (Common Vulnerability Scoring System) Funnel model as outlined in the course project slides:

CRITICAL SEVERITY (CVSS 9.0 - 10.0)	Public Network Firewall Disabled (Exposes open listening ports to local network attackers).
HIGH SEVERITY (CVSS 7.0 - 8.9)	Outdated Web Browser (Chrome v114) & Pending Cumulative Windows Security Patches.
MEDIUM SEVERITY (CVSS 4.0 - 6.9)	Disabled Account Lockout Policy & USB AutoRun feature enabled.
LOW SEVERITY (CVSS 0.1 - 3.9)	Microsoft Edge SmartScreen URL filtering feature turned off.

5. Audit Findings & Hardening Recommendations

The table below details all specific vulnerabilities identified during the audit alongside CVSS risk scoring and Blue Team remediation steps:

Finding ID	Identified Vulnerability	Category	CVSS Risk	Recommended Solution
VULN-01	Public Network Firewall turned off	Network	CRITICAL (9.1)	Enable Windows Defender Firewall for all network profiles via PowerShell.
VULN-02	Outdated Chrome & Adobe Reader	Software	HIGH (7.8)	Run winget upgrade --all to install current security patches.
VULN-03	Windows Updates paused (3 pending)	Patch Mgmt	HIGH (7.5)	Enable wuauserv service and apply pending cumulative update KB5029351.
VULN-04	No Account Lockout threshold	Password	MEDIUM (5.3)	Configure secpol.msc to lock account after 5 failed attempts for 15 mins.
VULN-05	USB AutoRun feature enabled	Endpoint	MEDIUM (4.6)	Disable AutoRun/AutoPlay via Group Policy to prevent BadUSB execution.
VULN-06	SmartScreen turned off in Edge	Web Safety	LOW (3.1)	Turn on SmartScreen in Microsoft Edge app settings.

6. Step-by-Step Remediation Action Plan

A. Enable Firewall across all profiles (Elevated PowerShell):

```
Set-NetFirewallProfile -Profile Domain,Private,Public -Enabled True
```

B. Enforce Automatic Patching & Software Upgrades:

```
Set-Service -Name "wuauserv" -StartupType Automatic
winget upgrade -all
```

7. Overall System Risk & Conclusion

OVERALL RATING: MEDIUM TO HIGH RISK

The target host machine (Dell XPS 15 - Windows 11 Pro) demonstrates solid core hardware security controls, such as active real-time Microsoft Defender protection and TPM 2.0. However, critical gaps in Network Security (Public Network Firewall disabled) and Patch Management (Pending OS patches and outdated browser binaries) leave the system vulnerable to network exploitation and drive-by malware attacks.

By executing the Blue Team hardening steps outlined in Section 6, the overall endpoint risk level will be reduced to LOW.
